

Topology-Correcting Differentiable Triangle-Soup Reconstruction via Persistent Homology

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Abstract

Resampling priors can steer *where* a differentiable triangle-soup reconstruction spends its triangle budget, but our companion study found their topological value is carried largely by spatial *width*, and that no prior shape repairs one-dimensional features: concentrated priors manufacture phantom handles, spread priors never beat a random control. We move topology from the allocation channel into the *objective*: a differentiable persistence loss $L = \text{photometric} + \lambda(t) L_{\text{topo}}$, where L_{topo} is an optimally-matched distance between the live surface’s persistence diagram and a fixed target diagram. Gradients flow through a pair-frozen backward pass: the matched birth/death simplices are re-expressed as closed-form circumradii of live sample positions — exact for the Gabriel simplices that constitute all significant pairs in practice — and a recruitment term restores the gradient that optimal matching provably lacks when a target feature is missing entirely. A single ratio knob ρ sets the loss scale by gradient-norm calibration; a curriculum proved unnecessary. On budget-starved synthetic scenes the loss is topology-specific where a norm-matched non-topological control is not: enclosed voids improve $4.0\text{--}7.9\times$ and — the channel’s distinguishing result — torus loops improve $2.4\times$ with zero phantom handles across seeds, while the control collapses a loop. The two channels compose: loss plus the best resampling prior reaches $7.6\times$ below baseline at the best Chamfer. Component counting (H0) is again matched by the generic control — a third channel agreeing that this class needs no topological information. All verdicts replicate, with no per-shape tuning, on three external genus-known meshes ($2.0\text{--}10.3\times$ at equal-or-better Chamfer). Prescription: correct topology in the loss, allocate with a wide prior, and combine.

1 Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [1], periodically resampling triangles under a fixed budget. Photometric losses are topologically blind: reconstructions that differ in genus, connectivity, or enclosed-void structure can be indistinguishable to Chamfer and Hausdorff distances while the bottleneck distance between persistence diagrams separates them by an order of magnitude. Our companion study [**TODO: cite companion paper (self-citation once venue/preprint id exists)**] attacked this through the *allocation* channel — biasing where the resampler spends triangles — and returned a mixed verdict: a topological prior helps enclosed voids, but most of the gain is reproduced by any equally-wide non-topological prior, and no prior shape repairs one-dimensional features: concentrating manufactures phantom handles, and spreading never beats a random control on loops.

This paper moves topology into the *objective*. We add a differentiable persistence term to the training loss, so gradients act on the birth/death values of the reconstruction’s topological features directly, rather than on where triangles respawn. Concretely:

- **A matched persistence loss with a pair-frozen backward (§3):** the forward pass is ordinary exact persistence (GUDHI alpha complex); the backward pass re-expresses each paired birth/death simplex as the closed-form circumradius of live surface samples. On our shapes, 100% of significant pairs are Gabriel-exact, so the frozen gradient is the true gradient almost everywhere.
- **A recruitment term for missing features:** optimal partial matching yields exactly zero gradient when a target feature has no live counterpart — the most important failure mode. Recruitment attaches the nearest live bar instead, restoring a descent direction; a controlled toy probes its known location-blindness (§5).
- **One calibrated knob:** λ is set by a gradient-norm ratio ρ against the photometric loss. The response is flat over a $10\times$ range of ρ , and a curriculum schedule adds nothing once calibrated — an ablation, not an assumption.
- **Channel-controlled evaluation:** every claim is tested against a norm-matched *non-topological* control acting through the identical channel (a repulsion on the same surface samples), inheriting the control discipline that made the allocation-channel study conclusive.

The loss is topology-specific for enclosed voids ($4.0\text{--}7.9\times$ error reduction at Chamfer parity) and, unlike every prior shape we tested, for loops ($2.4\times$, zero phantom handles across seeds). The two channels compose (best-of-both $7.6\times$), and component structure (H0) is once again matched by the generic control — three channels now agree that this class carries no exploitable topological signal.

2 Related work

Differentiable rasterization and soup pipelines. OpenDR [2], neural mesh rendering [3], SoftRas [4] and nvdiffrast [5] established gradient flow from pixels to geometry; Gaussian splatting [6] and triangle splatting [7] trade connectivity for speed. The pipeline we build on is DiffSoup [1], whose in-loop prune/respawn resampler is exactly the mechanism our companion study steered with importance priors; here the resampler stays untouched and the objective changes instead. Vertex-domain optimizers such as large-steps preconditioning [8] are complementary — they change *how* vertices move, we change *why*.

Persistence in learning objectives. Persistent homology [9] with its stability guarantees [10] has been used as a trainable signal via differentiable sorting of filtration values [11], for topology-preserving image segmentation [12, 13], point-cloud and shape optimization [14, 15], and diagram-metric optimization with convergence analysis [16]. Our backward pass is in this family — gradients through the filtration values of paired simplices — with three pipeline-specific ingredients: an *alpha-complex* pairing over $\alpha \times$ area surface samples of a live triangle soup (rather than image grids or raw clouds), a Gabriel-exactness gate that certifies the circumradius re-expression, and the recruitment term for missing target features, which pure optimal matching cannot see. **[TODO: verify-and-add 1–2 recent differentiable-PH refs if the venue expects them; do NOT cite unverified entries]**

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [17–19], and budgeted remeshing has a long lineage [20–24]. Our contribution is orthogonal: the budget mechanics stay DiffSoup’s; only the training signal becomes topology-aware.

3 Method: a matched persistence loss for triangle soups

3.1 Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary + $0.8 L_1 + 0.2 \text{SSIM}$) [1]. We add

$$L = L_{\text{photo}} + \lambda(t) L_{\text{topo}}(V), \quad (1)$$

where V are the soup’s vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed *target bundle* $\mathcal{T}$: the significant finite bars of the ground-truth shape’s diagram, computed once at the same sampling density M the loss will see (alpha birth/death values shift with point spacing, so density matching is a correctness requirement, not a convenience) and normalized by the target’s bounding-box diagonal, which is locked for all live evaluations.

3.2 Live diagram and pair-frozen backward

At each refresh we draw $M=2048$ surface samples $X_j = \sum_i w_{ij} V[F(f_j), i]$ with faces f_j drawn $\sigma(\alpha) \times \text{area}$ -weighted and barycentric weights w_{ij} frozen; X is recomputed from live V every step, so gradients reach vertices while the combinatorics stay fixed. The alpha complex of X yields persistence pairs (birth simplex, death simplex) per dimension. For a Gabriel simplex the alpha filtration value is its squared circumradius, so each paired bar’s coordinates are closed-form functions of a few sample positions — half edge length (H0 deaths), triangle circumradius (H1 deaths, H2 births), tetrahedron circumradius (H2 deaths) — all exactly differentiable. We verify, per pairing, that the re-expressed value matches the filtration value (relative 10^{-6}); non-Gabriel simplices would contribute value without gradient, but on all six benchmark shapes 100% of significant pairs are Gabriel-exact. The pairing is treated as locally constant — the standard pair-frozen device [11, 16] — and refreshed every $K=10$ steps and, necessarily, immediately after every resampling event, which replaces the vertex and face tensors wholesale.

3.3 Matching, diagonal pressure, and recruitment

Let D be the live significant bars and $\mathcal{T}$ the target bars of one dimension. An optimal partial matching (Hungarian on the standard augmented cost; equivalently W_2) splits bars into matched pairs, unmatched live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j) \text{ matched}} [(b_i - b_j^*)^2 + (d_i - d_j^*)^2] + \sum_{i \text{ unmatched}} \left(\frac{d_i - b_i}{2} \right)^2 + L_{\text{recruit}}. \quad (2)$$

The first term pulls matched features to their target coordinates; the second pushes spurious features to the diagonal. The third exists because optimal matching has a blind spot: when a target feature is entirely missing, the cheapest transport diagonalizes both sides and the gradient is exactly zero — precisely in the most important failure case. Recruitment attaches each unmatched target bar to the nearest not-yet-matched live bar (sub-threshold bars allowed) and pulls it toward the target. This deliberately departs from a metric-faithful W_2 ; §5 probes its known pathology (location-blindness) directly.

3.4 Calibration instead of curriculum

$\lambda(t)$ ramps linearly from 20% to 50% of training, with peak set once by a gradient-norm ratio at the first active step: $\lambda_{\text{peak}} = \rho \|\nabla_V L_{\text{photo}}\| / \|\nabla_V L_{\text{topo}}\|$. The single knob ρ is dimensionless; $\rho=0.1$ throughout. The tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, and an uncalibrated-schedule ablation (constant λ from step one) matches the ramped version — calibration, not scheduling, is what makes the knob safe.

3.5 Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: $+$ a short-range repulsion on the *same* frozen samples, calibrated with the *same* ρ — a norm-matched non-topological control through the identical channel (C2g: a gentler radius). C3: C1 without the ramp. C5: C1 combined with the best allocation-channel prior of the companion study (the spread topological resampling field, B4). The hook adds one loss line to the training loop; with the flag off, no new code executes, and a $\rho=0$ run is divergence-equivalent to baseline re-runs under CUDA nondeterminism.

4 Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (the loader’s standard every-8th test hold-out leaves 21–63 training views): sphere and cube (void class H2, $\beta=(1, 0, 1)$, budget $N=1200$), torus (loop class H1, $(1, 2, 1)$, $N=700$), two spheres (component class H0, $(2, 0, 2)$, $N=700$), and genus-2 double torus as a stretch case $((1, 4, 1)$, $N=2000$). Budgets are the companion study’s headroom points — tight enough that the baseline’s topology is imperfect, so there is something to win. Three external, genus-known, watertight meshes probe generality beyond the analytic family (§5.3): *spot* (genus 0) and *bob* (genus 1) from the Crane model repository (CC0), and the *fandisk* CAD benchmark (genus 0) — rendered with the same pipeline (48 views, 200 px), class budgets carried over unchanged ($N=1200$ voids, $N=700$ loops), and a seed-0 C0 pilot confirming baseline imperfection (tail bottleneck .043–.054, i.e. inside the analytic shapes’ band).

Protocol. 2,500 steps; resampling every 100 steps (untouched); $\rho=0.1$, ramp 0.2:0.5, refresh $K=10$, $M=2048$; decisive shapes (sphere, torus) run 5 seeds with the full condition set (C0/C1/C2/C3/C5), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2), double torus 2 seeds. The torus restricts the loss to H1: its own void bar is sub-threshold at $M=2048$, and a loss must not act on features it cannot reliably measure at its sampling density. For the same reason the double torus targets its two measurable loops (of four) — matching what the reconstruction itself can express at any feasible budget. The generality trio runs C0/C1/C2 with three seeds; bob, unlike the thin torus, has both its loop pair *and* its void significant at $M=2048$ (fat tube), so it runs the full unrestricted bundle.

Metrics and verdict. Primary: bottleneck distance to the target diagram in the shape’s discriminating dimension, tail-averaged over the last trajectory dumps, seed-averaged (20k evaluation samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. *Verdict rule:* C1 must beat C0 **and** every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not a topological win. CUDA training is not bit-reproducible (baseline re-runs diverge at the first resample), so no bit-level claims are made anywhere.

Table 1: Tail bottleneck to target (lower is better); $\times$ = reduction vs C0. Verdict requires C1 < C0 and C1 < every control at Chamfer parity.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior
sphere (H2)	.0623 $\pm$.0063	.0156$\pm$.0013 (4.0 $\times$)	.1372 (worse)	.0151 $\pm$.0004	.0082$\pm$.0007 (7.6 $\times$)
cube (H2)	.0582 $\pm$.0047	.0074$\pm$.0011 (7.9 $\times$)	.1346 (β_2 1 $\rightarrow$ 0)	—	—
torus (H1)	.0424 $\pm$.0031	.0180$\pm$.0016 (2.4 $\times$)	.0457 (β_1 2 $\rightarrow$ 1)	.0155 $\pm$.0014	.0133$\pm$.0004 (3.2 $\times$)
two_spheres (H0)	.0065 $\pm$.0008	.0007 $\pm$.0002 (9.3 $\times$)	.0008 (ties C1)	—	—
double_torus (H1)	.0264 $\pm$.0000	.0264 $\pm$.0000 (saturated)	.0264 $\pm$.0000	—	—

5 Results

5.1 Gate: the loss alone repairs topology

Before integration, the loss (no photometric term, Adam directly on point positions) must repair defective clouds. It does, on all four probes (Fig. 1): a punctured sphere’s void birth is pulled to target (8.8 $\times$ bottleneck reduction; the hole closes), a spurious handle is pushed to the diagonal and dies (H1 bar eliminated; β restored), mis-spaced components are pulled to the target separation (1229 $\times$), and — the recruitment probe — a bridged merge with *no* significant live H0 bar is severed (450 $\times$, β_0 1 $\rightarrow$ 2). The predicted location-blind pathology did not materialize, for a mechanism reason worth stating: tearing a 2-manifold shell cannot grow a component bar past the local detour scale (the tug is abandoned), while cutting the 1-D bridge raises the recruited bar’s death monotonically toward target — the runaway concentrates on the only profitable direction.

5.2 The C-matrix

Table 1 reports tail bottleneck (mean $\pm$ sd over seeds) in each shape’s discriminating dimension; Chamfer in the text where it matters; Fig. 2 shows the full training trajectories for the two decisive shapes.

Voids (H2): topology-specific, large. Sphere 4.0 $\times$ (16.2 σ Welch vs C0), cube 7.9 $\times$ (18.5 σ), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is not merely weaker — it is *destructive*: it pins the void’s death at its repulsion equilibrium (bottleneck 2.2 $\times$ worse than baseline, identical across seeds) and on the cube erases the void outright. Nor is this an artifact of control strength: a gentle variant at half the push radius still lands 1.6 $\times$ *worse* than baseline (.0972 vs .0623) — the sphere ordering is C1 < C2g < C0 < C2, so no tested repulsion strength reproduces any of the gain (Fig. 3).

Loops (H1): the channel’s distinguishing result. In the companion study no prior shape won this class — concentration manufactured phantom handles; spreading never beat a random control. The loss cuts torus H1 error 2.4 $\times$ with *zero phantom handles in all five seeds* (significant-count trajectories flat at the true $\beta_1=2$, Fig. 4), while the control collapses one loop. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not.

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error 9.3 $\times$, but the generic control matches it on the diagram metric (.0008 vs .0007); the topological increment appears only in Chamfer (0.54 vs 1.94). Together with

Table 2: Generality wave: tail bottleneck to target (mean $\pm$ sd, 3 seeds); $\times$ = reduction vs C0; σ = Welch. Same verdict rule as Table 1.

shape (dim)	C0	C1 loss	C2 control	verdict
spot (H2)	.0521 $\pm$.0081	.0237$\pm$.0000 (2.2 $\times$)	.0652 (β_2 1 $\rightarrow$ 0)	PASS (6.1 σ)
bob (H1)	.0426 $\pm$.0012	.0208$\pm$.0008 (2.0 $\times$)	.0437 (β_1 2 $\rightarrow$ 1)	PASS (26.1 σ)
fandisk (H2)	.0538 $\pm$.0022	.0052$\pm$.0008 (10.3 $\times$)	.0571 (β_2 1 $\rightarrow$ 0)	PASS (35.7 σ)

the allocation-channel results this is now a statement about the feature class: component structure is helped by any spreading pressure, not by topological information.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 (7.6 $\times$ vs C0, $\approx$ 2 $\times$ vs C1 alone, $\approx$ 4.6 $\times$ below the prior alone), torus .0133 — with the best Chamfer as well. Allocation and value-tuning are complementary mechanisms, not substitutes.

Stretch case (genus 2): saturated, honestly. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly .0264 (sd 0 over seeds), and Wasserstein is equally pinned (.0527–.0528): the two tube loops are unexpressable at feasible budgets (replicating the companion study’s floor), and they dominate both diagram metrics regardless of guidance. The loss adds no phantoms there (β_1 -significant count = 2 in every run) and its Chamfer is marginally best; the case measures the *representation’s* ceiling, not the channel’s.

Ablations. ρ -response flat over {0.03, 0.1, 0.3} (.0181/.0172/.0177 on sphere); C3 (no curriculum) $\approx$ C1 on both decisive shapes. Overhead: $\approx$ 45–55 s per 2,500-step run (refresh 174 ms at $M=2048$, every 10 steps).

5.3 Generality: external genus-known meshes

Everything above is on one analytic scene family. We therefore repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ , ramp and sampling unchanged — on three external watertight meshes of known genus (§4): *spot* (genus 0, organic), *bob* (genus 1, organic), and the *fandisk* CAD benchmark (genus 0). No per-shape tuning of any kind: the headroom budgets transfer from the feature class, and bob runs the full unrestricted bundle. All three **pass** the verdict rule at equal-or-better Chamfer (Table 2, Fig. 5), and the control’s failure modes replicate exactly: repulsion erases the void on both H2 shapes (β_2 1 $\rightarrow$ 0, all seeds) and collapses a loop on the genus-1 shape (β_1 2 $\rightarrow$ 1, all seeds).

Three observations. *The H1 claim survives the family change:* bob halves loop error with zero phantom handles in every seed (26.1 σ), replicating torus. *Sharp-featured CAD benefits most:* fandisk’s 10.3 $\times$ exceeds every analytic-family effect. *Complex organic geometry exposes a count-level limit:* spot’s *baseline* manufactures a phantom void in two of three seeds (final #sig H2 = 2, 1, 2); the loss clears it in two seeds but retains one spurious bar in the third (1, 2, 1) — value repair is reliable, count repair on sub-budget organic detail (legs, horns at $N=1200$) is an improvement, not a guarantee. Spot’s C1 tail is also pinned across seeds (sd 0 at .0237), suggesting a geometric floor of the same kind as the double torus, far milder in degree.

6 Mechanism and discussion

Sparse tugs vs. budget allocation. Each significant pair back-propagates through at most a handful of sampled points, so the loss delivers *precise but sparse* pulls, iterated by re-pairing — the toy probes show surgery “walking” around a defect a few points at a time. The resampling prior is the opposite: broad, indirect, budget-limited. The results sort cleanly along this axis: 1-D loops, which the prior could only over- or under-tessellate, respond to precise value pressure (H1 win); wide 2-D shells benefit from both (H2: prior helps, loss helps more, composition best); 0-D component structure needs neither — any spreading pressure suffices (H0 null, three channels).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface; the void’s death time pins at that equilibrium (identical across seeds) and thin structures collapse (β_2 loss on the cube, β_1 loss on the torus). This sharpens the control argument: spending the same gradient budget through the same channel *without* topological information is not neutral — it is harmful — so C1’s wins cannot be an artifact of extra regularization.

Measurability at the loss’s density. A persistence loss can only enforce features that clear the significance floor at its sampling density M : the torus void and the double torus’s two tube loops do not, at $M=2048$. We treat this as a principle rather than a nuisance: the target bundle is built (and audited) at the same M , features below the floor are excluded from the objective, and the double-torus stretch case is evaluated against its two measurable loops — which is also all the reconstruction itself can express at feasible budgets. Raising M buys resolution at linear-ish persistence cost; the trade is explicit and logged at bundle-build time.

Recruitment is landscape-guided in practice. Its location blindness — pull the nearest bar anywhere — did not misfire on our geometry because unproductive tears self-extinguish (their bar’s death saturates at the local detour scale) while productive cuts run away toward the target. The caveat stands for target features larger than any achievable detour; the probe and its mechanism are reported so the term is used with eyes open.

Prescription. Correct topology in the *loss*; allocate budget with a *wide* prior; combine for the best of both. Curriculum schedules are unnecessary once the loss scale is calibrated against the photometric gradient.

7 Limitations

Synthetic, closed, single-machine. All quantitative claims are on synthetic renders of closed surfaces with known topology, on one GPU. The scene family is analytic plus three external genus-known meshes (§5.3) — still short of a public shape-set benchmark — and a real (open-surface, clinical) demonstration remains future work; open surfaces add a genuine research obstacle — scan boundaries contaminate H1, and no clean ground-truth diagram exists.

Density floor. The loss cannot see features below the significance threshold at its sampling density (torus void; two of four double-torus loops at $M=2048$). The bundle audit makes this visible, not invisible — but it is a real ceiling on what the objective can enforce.

Fixed target diagram. The target bundle presumes known ground-truth topology (true in reconstruction-from-scans settings where a prior scan or template exists; not in blind capture). Template-free variants (e.g., losses on diagram *statistics*) are untested here.

Control strength. The repulsion control is aggressive (it destroys topology at the calibrated magnitude); a gentler variant at half the radius is still $1.6\times$ worse than baseline (sphere: .0972 vs .0623), so the verdicts do not hinge on control tuning — but control *design* for loss-channel studies deserves more variety than one family at two strengths.

Nondeterminism. DiffSoup’s CUDA training is not bit-reproducible — even baseline re-runs diverge in resampling decisions — so all statements are seed-averaged; per-run reproductions will differ in detail.

Deferred arms. Curvature-weighted spatial masking of the loss (C4) and the dental showcase are designed but not run; both are gated on a small weight-mask extension to the loss state.

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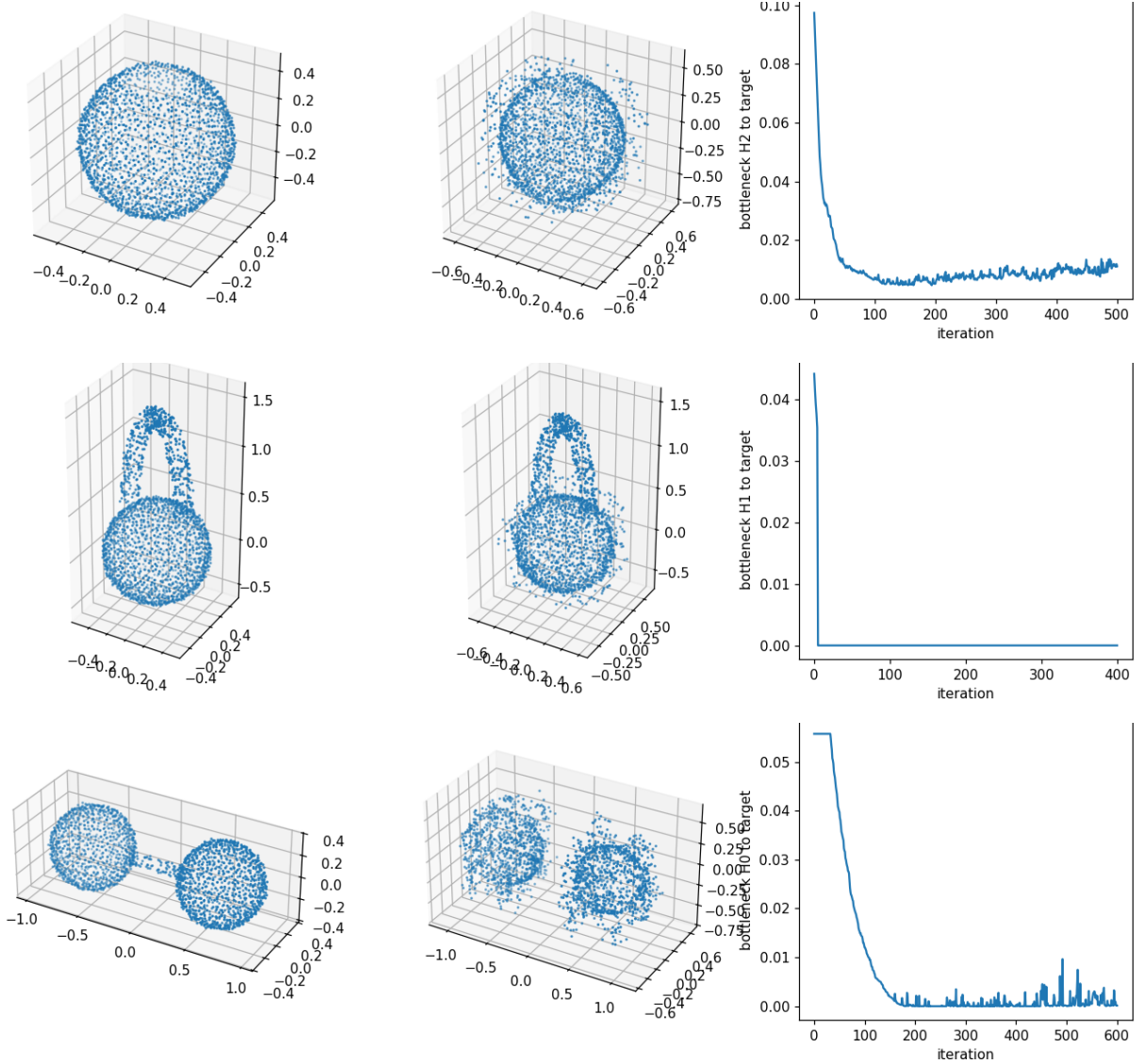


Figure 1: The stage-3a gate: the loss alone (Adam on raw point positions, no photometric term) repairs defective clouds. Each row shows the initial cloud, the final cloud, and the bottleneck-to-target trajectory in the discriminating dimension — one row per mechanism of the matched loss (§3). *Top* (T1): punctured sphere; the matched birth term pulls the void closed ($8.8\times$ reduction). *Middle* (T2): spurious handle; the diagonal term kills the extra H1 bar outright. *Bottom* (T4): bridged merge with no significant live H0 bar; recruitment severs the bridge ($450\times$, β_0 1 $\rightarrow$ 2) — the case where plain optimal matching provably has zero gradient. The mis-spaced components probe (T3, $1229\times$) is omitted for space.

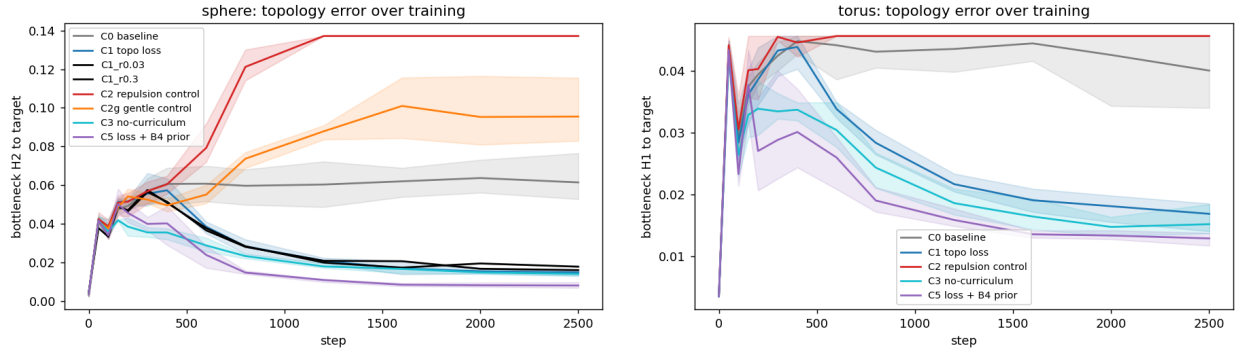


Figure 2: Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min-max). *Left*, sphere (H2): every loss arm — C1, C3, C5, and both ρ ablation variants — converges to a fraction of baseline, while the norm-matched repulsion control C2 is *destructive* and even its gentle variant C2g ends worse than baseline. *Right*, torus (H1), the class no resampling prior won: same ordering, with C5 (loss + spread prior) best.

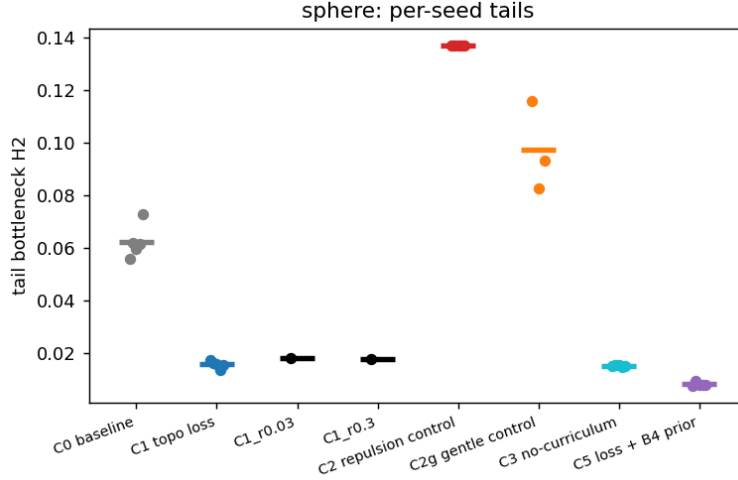


Figure 3: Sphere, per-seed tail bottlenecks (dots) with arm means (bars). Every loss arm sits far below baseline with tight seed spread; the two control strengths bracket baseline from *above*, so no tested repulsion strength reproduces any of the gain; the ρ ablation arms are indistinguishable from $\rho=0.1$.

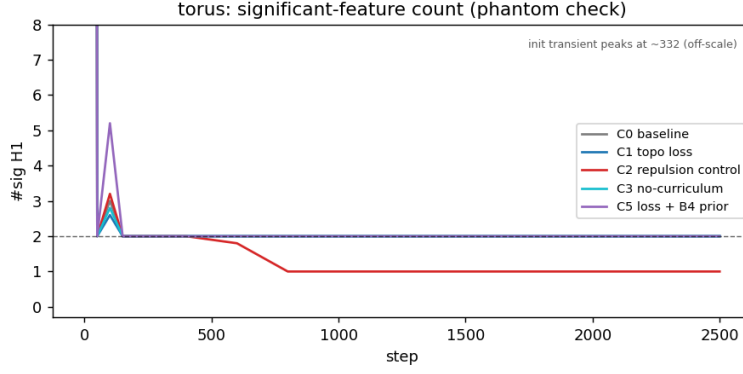


Figure 4: Torus phantom check: significant-H1 count over training (seed means; dashed line: true $\beta_1=2$). After a brief resampling transient near step 100, every loss arm sits exactly on two significant loops for the rest of training — zero phantom handles — while the repulsion control (C2) collapses a loop by step ~ 800 and never recovers it. All arms share the off-scale random-init transient (annotated).

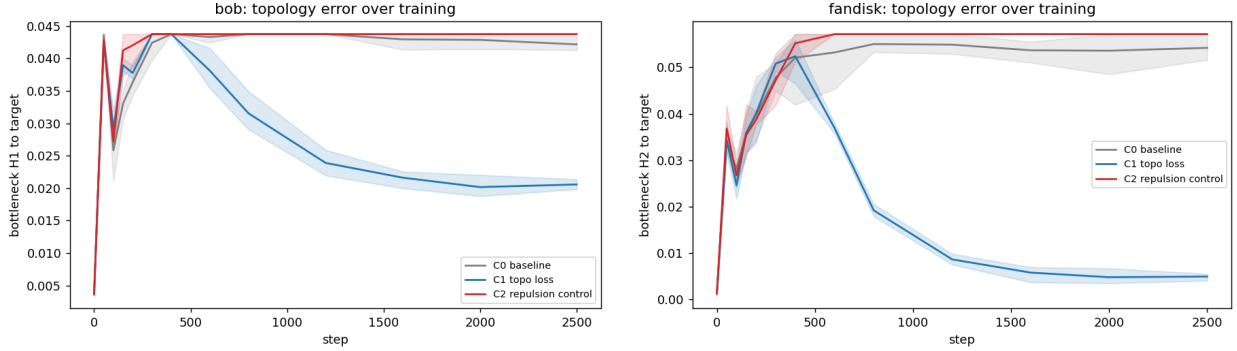


Figure 5: Generality trajectories (line = seed mean, band = seed min-max). *Left*: bob (genus 1) — the H1 result off the analytic family: the loss halves the loop error with $\# \text{sig H1} = 2$ in every seed, while the control rides the baseline after collapsing a loop. *Right*: fandisk (CAD) — the largest effect measured in this study ($10.3\times$).